

JOSE EMMANUEL RODRIGUEZ RIOS

(407) 729-9500 | jerr.ccom@gmail.com | github.com/joserico00 | linkedin.com/in/jose-rodriguez-rios/ | joserico00.github.io

Computer Scientist and Cybersecurity Software Developer

Highly motivated to work with AI, cybersecurity, reverse engineering, and data science.

Seeking a full-time position; also open to co-op and internship roles.

EDUCATION

Ph.D., Software Engineering, UPR – Mayagüez Campus Aug 2026 – May 2030 (expected)

M.S., Computer Science, UPR – Río Piedras Campus | GPA: 4.0 Aug 2022 – May 2025

B.S., Computer Science, UPR – Río Piedras Campus | Minor in Cybersecurity | GPA: 3.29 Aug 2015 – Dec 2020

TECHNICAL AND SOFT SKILLS

- **Programming languages:** Python, C, C++, Assembly, Bash
- **Data science:** pandas, NumPy, scikit-learn, LightGBM, Matplotlib, seaborn, Jupyter, Excel/CSV reports, Dash
- **Machine learning:** PyTorch, TensorFlow/Keras, Hugging Face transformers, CNNs, object detection, LLMs (Ollama)
- **HPC and cloud:** SLURM, GPU training on Perlmutter, Polaris, and Aurora; AWS SageMaker
- **Vulnerability assessment:** Nmap, OpenVAS, Vulners, vulscan, Nikto, Nuclei, NVD API
- **Security testing:** Metasploit, Burp Suite, Wireshark, Scapy, Hashcat, Hydra, Gobuster, John the Ripper
- **Reverse engineering and forensics:** Ghidra, BSim, GDB, TestDisk, Foremost, Autopsy, SIEM/IDS
- **Systems and DevOps:** Linux, Docker, GitLab CI/CD, Git, Elasticsearch, Flask, Playwright, REST APIs, OpenVPN, WireGuard
- **Soft Skills:** Bilingual (English/Spanish); teamwork; leadership; Agile/Scrum; mentoring; technical writing; fast learner

EXPERIENCE

Research Intern, Argonne National Laboratory, Lemont, IL Jun 2026 – Aug 2026

- Built **DNA classifiers** with **PyTorch**, **Hugging Face**, and **DNABERT** on **Perlmutter**, **Polaris**, and **Aurora**. Compared **6-mer/BPE** tokenization: **81%/80%** accuracy and **195/34 tokens** per sequence.
- Led **10 students** through a **10–12-hour** project with **Jupyter/SLURM** lessons, **CPU/GPU** experiments, and final presentations; the team fine-tuned **DNABERT** to **91.7%** evaluation accuracy on CTCF data.

Graduate Research and Teaching Assistant, UPR Río Piedras, San Juan, PR Aug 2022 – May 2025

- Developed **NVSRCO**, an open-source vulnerability scanner using **Python**, **Nmap**, and **OpenVAS**, with **CSV** reporting, a **Dash** dashboard, and **WireGuard** links to remote networks.
- Led OS, Computer Architecture, and Cybersecurity labs with **Linux**, **Ghidra**, **Wireshark**, and **Scapy**; authored **CTF** challenges in reverse engineering, cryptography, web, and forensics for about **60** Puerto Rico high-school students.

DevOps Intern, Lawrence Livermore National Laboratory, Livermore, CA Jun 2024 – Aug 2024

- Built a containerized **Elasticsearch** store for **Ghidra BSim** binary features so researchers can query malware-like samples at scale, with **GitLab CI/CD** for deployment.
- Deployed a **Docker**-hosted **Ollama (Llama 3)** service for the **PaperQA** PDF-reading application.

Volunteer Software Developer, Abeyno Coop solar cooperative, Salinas, PR 2024 – Present

- Built **Solar Fleet Reporter**: a **1,700-line Streamlit** app with **88 automated checks** for solar data collection, net-metering calculations, and **CSV/Excel** reports and printable statements.

Cybersecurity Intern, GM Sectech, San Juan, PR Jun 2022 – Aug 2022

- Built a **Python/Twilio** smishing (SMS phishing) simulator for awareness training; collaborated with penetration testers on credential-theft and domain-compromise scenarios; trained on **Metasploit**, **Hashcat**, and **Burp Suite**.

AI AND CYBERSECURITY PROJECTS

Malware Image Classification

- Built a **TensorFlow CNN** to classify **9,401** malware images across **25 families** in the **Maling** dataset.
- Reached **97.7%** validation accuracy on malware image classification.

Malicious Network Traffic Detection

- Built a **TensorFlow** pipeline for **IoT-23** logs with capture-aware splits and training-only preprocessing.
- Compared a class-weighted **dense neural network** with a **logistic-regression** baseline.

AWARDS

- **Lockheed Martin AI/ML Hackathon (2026): 3rd of 25 teams**; object-detection ensemble, **0.8441 test mAP@50**.
- **CAHSI Data Analytics Challenge (2026): 1st of 21 teams**; **LightGBM**, network-intrusion classifier, **0.9428 Macro-F1**.
- **CAHSI Cyber Hackathon (2026): 2nd place**, Advanced Category; penetration testing, reverse engineering, forensics.
- **BSides Puerto Rico CTF (2024): 1st place of 30 teams**; reverse engineering, cryptography, exploitation.
- **Bloomberg HSI Tech Summit, New York (2024): 1 of 29 students** selected from **400+ applicants**.
- **Bloomberg Summer Accelerator (2024):** invited to the **6-week program** of mentored coding challenges.
- **Bloomberg Hackathon (2022): 5th place**; built **Seniory**, a **Python/SQLite** app matching seniors with volunteers.
- **Tracer Fire Blue Team, ACSAC (2019): 2nd place team of 40 participants**; SIEM/IDS, Autopsy, Ghidra.